Learning Intrinsic Dimension of Complex Large Datasets

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Outline

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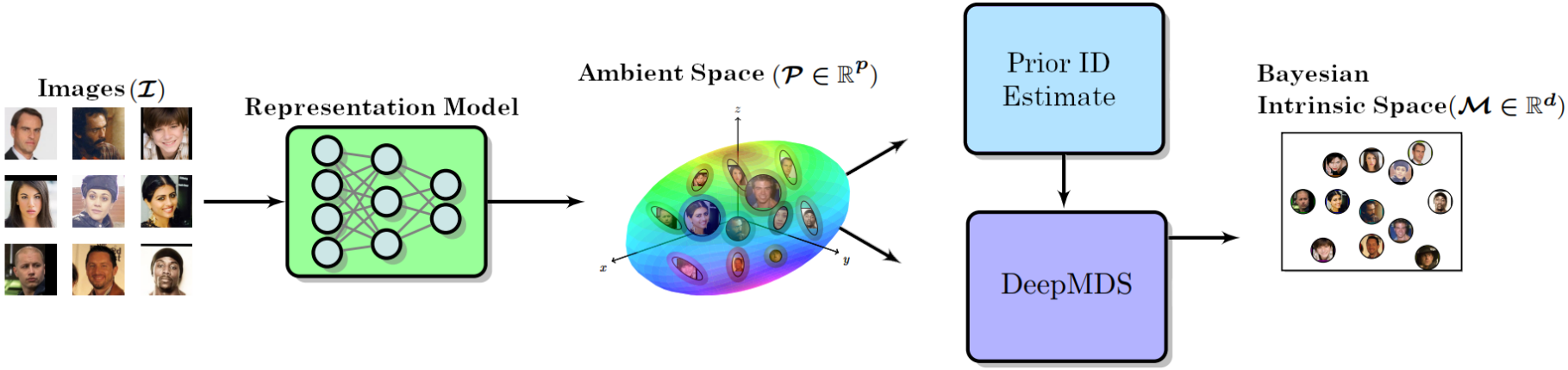
Introduction

Intrinsic Dimension

R. S. Bennent (1965)

The dimensionality of a collection of signals (vectors) is defined as equal to the number of free parameters required in a hypothetical signal generator capable of producing a close approximation to each signal in the collection.

Definition



[Gong, Boddeti, and Jain (2019)]

Intrinsic dimensionality (ID)

The number of parameters d of the lowest d-dimensional space such that some function of a p-dimensional space is mostly preserved.

Goals

1. Verify DeepMDS Results

Reproduce the results from [Gong, Boddeti, and Jain (2019)] by reducing 512-dimension image representations down to 19 dimensions with a loss of mAP of 6

2. Run BNN at Scale

Train the Bayesian Neural Network from [Liu, Bhattacharya, and Maiti (2021)] using 130,000 image representation, and compare the ID and mAP to DeepMDS.

3. Add TwoNN as Bayesian Prior

Estimate the intrinsic dimension of 130,000 ImageNet images using the TwoNN approach.

Methods

Variational Inference

The Setup

Loss (Multidimensional Scaling):

$$Z_i = f_\theta(X_i),$$

where $f_\theta(X_i)$ is a neural network.

$$d(X_i, X_j) = d(Z_i, Z_j) + \epsilon_{ij}$$

where $\epsilon_{ij} \sim N(0, \sigma^2)$.

$$\log L(\theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \Big[\|X_i - X_j\|_2 - \|Z_i - Z_j\|_2 \Big]^2 + \frac{1}{2} \log(2\pi\sigma^2)$$

where X_i is the ith input image representation, and Z_i is the associated reduced image representation.
For simplicity we assume $\sigma = 1$.

Variational Inference

Posterior

$$p(\theta|data) \propto e^{\log L(\theta)+log(p(\theta))}$$

where $p(\theta) \sim N(\mathbf{0}, \sigma_0^2\mathbf{I})$ is a Gaussian prior.

Variational Family:

$$q(\theta) \sim N(\mu, \Sigma)$$

where μ, Σ are estimated parameters.

Variational Posterior

We choose $q(\theta)$ such that the Kullback-Leibler divergence

$$KL(q(\theta), p(\theta|data))$$

is minimized. Or equivalently minimize

$$E_q(\log q(\theta) - \log p(\theta|data)) = E_q(\log q(\theta) - \log L(\theta) - \log p(\theta)) + C$$

where C the proportionality constant in $p(\theta|data)$ can be ignored.
So finally we minimize the following

$$-E_q(\log L(\theta)) + KL(q, p)$$

Bayesian Parameters

Weights

$$W^{(l)} \in \mathbb{R}^{p^{(l-1)} \times p^{(l)}} \sim N(vec(\mu_W), log(1 + e^{vec(\rho_W)})), l = 1, 2, ..., L$$

where $\mu_w \in \mathbb{R}^{p^{(l-1)} \times p^{(l)}}$ and $\rho_w \in \mathbb{R}^{p^{(l-1)} \times p^{(l)}}$ are trainable parameters.

Bias

$$b^{(l)} \in \mathbb{R}^{p^{(l)}} \sim N(\mu_V, log(1 + e^{\rho_V})), l = 1, 2, ..., L$$

where $\mu_v \in \mathbb{R}^{p^{(l)}}$ and $\rho_v \in \mathbb{R}^{p^{(l)}}$ are trainable parameters.

$p^{(l)} \in \mathbb{N}$ be the dimension of layer l.
When evaluated under the variation inference framework, $W^{(l)} = \mu_w + zlog(1 + e^{\rho_w})$ where $z \sim N(0, I)$

Layers

Linear Layer

$$a^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}, l = 1, 4, 7, 11$$

BatchNorm Layer

$$a^{(l)} = W^{(l)} \frac{a^{(l-1)} - \mu}{\sqrt{\Sigma}} + b^{(l)}, l = 2, 5, 8, 12$$

where $\mu = \mathbb{E}(a^{(l-1)})$ and $\Sigma = Var(a^{(l-1)})$

Activation Layer

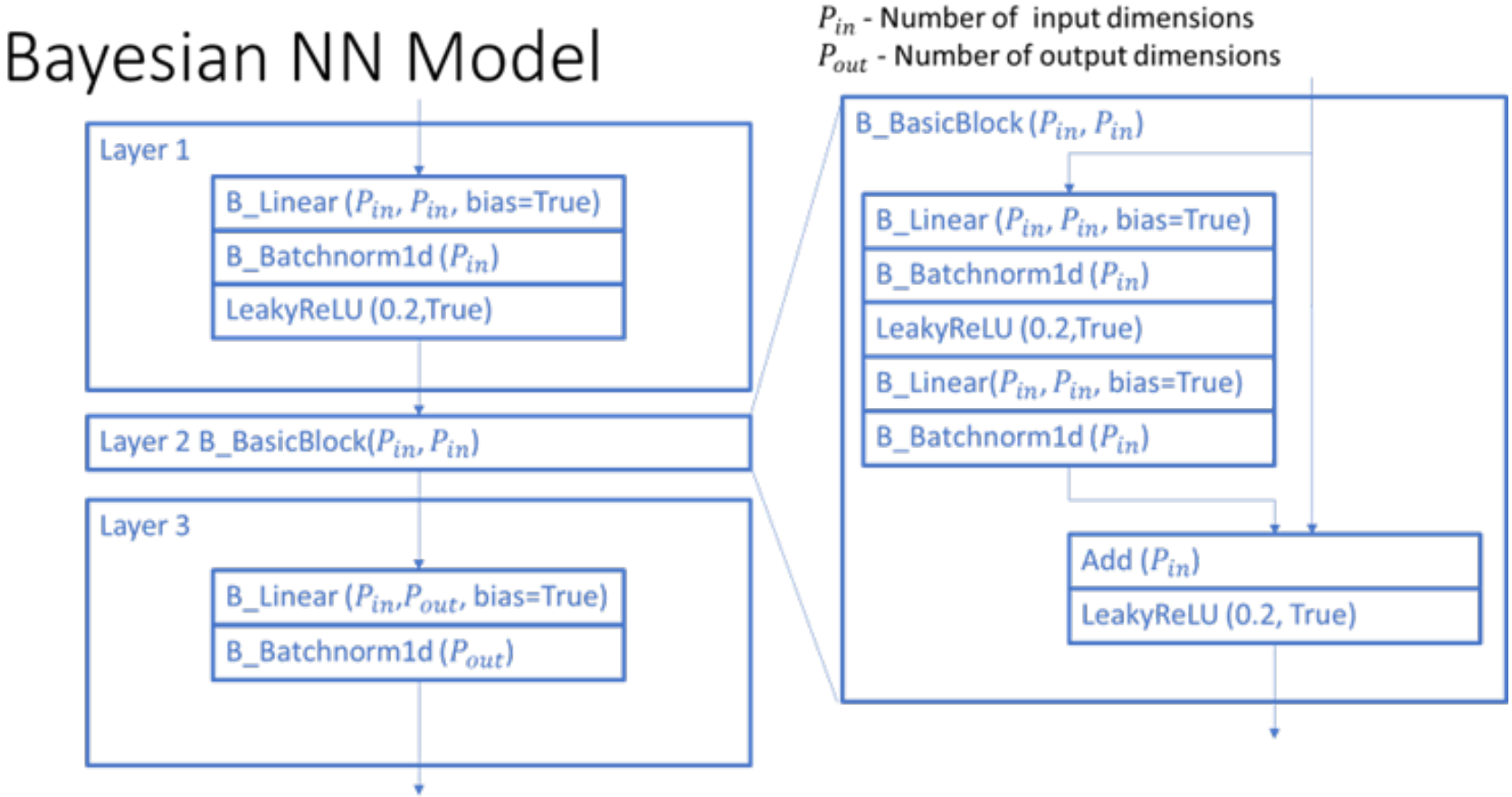
$$a^{(l)} = f^{(l)}_{LReLU}(a^{(l)}), l = 3, 6, 10$$

LeakyReLU activation vector function.

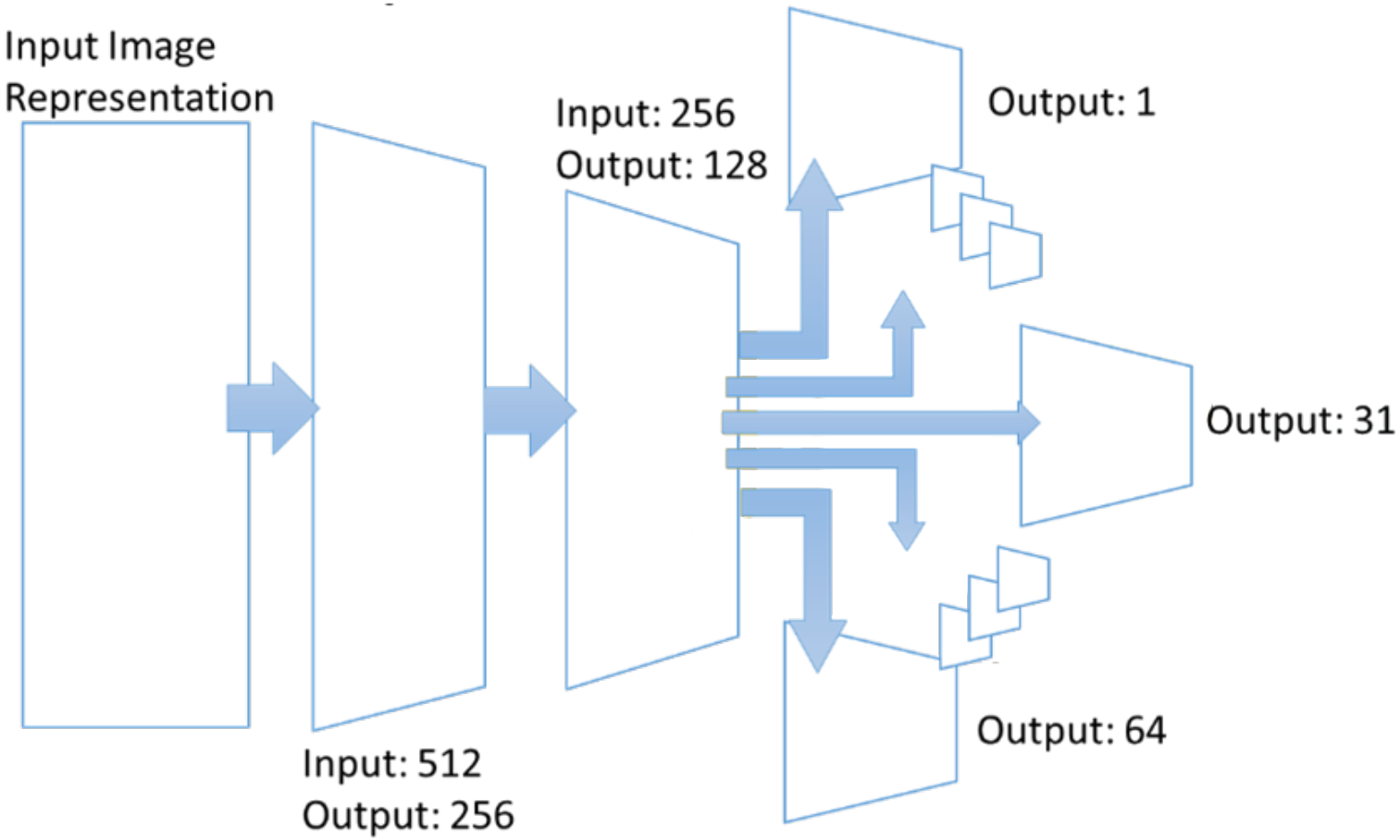
Skip Layer

$$a^{(l)} = a^{(l-1)} + a^{(l-6)}, l = 9$$

Bayesian Neural Network Architecture



Ensemble of BNNs



Loss Function (Estimated Lower Bound (ELBO))

Likelihood (Multidimensional scaling)

$$LL = \sum_{i=1}^{n} \sum_{j=1}^{n} \Big[\|X_i - X_j\|_2 - \|Z_i - Z_j\|_2 \Big]^2$$

where X_i is the ith input image representation, and Z_i is the associated reduced image representation.

KL Divergence

$$KL = \sum_{l=1}^{L} \left[\sum_{i=1}^{p^{(l)}} \sum_{j=1}^{p^{(l-1)}} KL(W_{ij}^{(l)} || \phi_{\sigma^2}) + \sum_{i=1}^{p^{(l)}} KL(b_i^{(l)} || \phi_{\sigma^2}) \right],$$

where $W_{ij}^{(l)}$ is the ij^{th} weight random parameter on layer l, $b_i^{(l)}$ is the ith bias random parameter on layer l, and ϕ_{σ^2} is $N(0, \sigma^2)$.

Algorithm

Initialization

Let μ_{i0} be randomly generated from $uniform(-.6, .6)$, and $\Sigma_{ii} = log(1 + e^{-6})$.

Train (AdamW optimization) [Loshchilov and Hutter (2017)]

Let $\beta_1 = 0.9, \beta_2 = 0.999$, weight_decay $\lambda = 1e - 4$, learning rate $\gamma_0 = 3e - 4, \epsilon = 1e - 8$, $m_0 = 0$, $v_0 = 0$
For t in $1, 2, \ldots$ apply update rule until convergence,

$$\theta_t = \theta_{t-1} - \gamma_t \lambda \theta_{t-1} - \gamma_t \frac{m_t}{\sqrt{v_t} + \epsilon}$$

where

$$m_t = \frac{\beta_1 m_{t-1} + (1 - \beta_1)\nabla f(\theta_{(t-1)})}{1 - \beta_1^t}, v_t = \frac{\beta_1 v_{t-1} + (1 - \beta_2)\nabla f(\theta_{t-1})^2}{1 - \beta_2^t},$$

Step size (Cosine Annealing Learning Rate)

$$\gamma_t = \gamma_{min} + \frac{1}{2}(\gamma_{max} - \gamma_{min})(1 + cos(t/t_{max}))$$

Accuracy Calculations

Mean Average Precision (mAP)

$$mAP = \frac{1}{100} \sum_{i=1}^{100} AP_i$$

Where the AP_i is the average precision for the ith category.

Average Precision (AP)

$$AP = \sum_{j=1}^{n} [P(j)\Delta r(j)]$$

where $P(j)$ is the precision, and $\Delta r(i)$ is the change in recall.

Precision and Recall

$$P(j) = TP(k_j)/(TP(k_j) + FP(k_j))$$

$$\Delta r(j) = \frac{TP(k_j)}{TP(k_j) + FN(k_j)} - \frac{TP(k_{j-1})}{TP(k_{j-1}) + FN(k_{j-1})}$$

where k_j is the jth decision threshold.

Estimate Intrinsic Dimension

Two NN Approach[Denti (2022)]

Consider Data $X \in \mathbb{R}^{n \times p}$ where X_i is i^{th} row vector.
Let $D \in \mathbb{R}^{n \times n}$ be the distance matrix s.t. $d_{ij} = \|X_i - X_j\|_2$.
Let $r_{i,1}, r_{i,2}$ be the two smallest non-zero values for i^{th} row of D.
Let $\mu_i = r_{i,2}/r_{i,1}$,
then,

$$\mu_i \sim Pareto(1, d), i = 1, 2, ..., n,$$

where shape parameter d is the intrinsic dimension.

Verification of Pareto Shape Parameter

Pareto Shape Estimation (MLE)

Consider minimization with respect to d of the log-likelihood of a $\mu_i \sim Pareto(1, d)$.

$$\min_d \ell(d, \mu) = \min_d ln \prod_{i=1}^{n} \frac{d}{x\mu_i^{d+1}} = \min_d nln(d) - (d+1)\sum_{i=1}^{n} ln(\mu_i)$$

$$\implies \frac{\partial \ell}{\partial d} = \frac{n}{d} - \sum_{i=1}^{n} ln(\mu_i) = 0 \implies \hat{d} = \frac{n}{\sum_{i=1}^{n} ln(\mu_i)}$$

According to [Rytgaard (1990)], $\mathbb{E}(\hat{d}) = \frac{n}{(1-n)}d$. We correct for the bias and verify what is described in [Denti (2022)]. Thus the MLE of d is,

$$\hat{d} = \frac{n-1}{\sum_{i=1}^{n} ln(\mu_i)}$$

Derivation of Posterior Distribution of Intrinsic Dimension

Bayesian Posterior Calculation

$$p(d|\mu) \propto p(\mu|d)p(d)$$

where $\mu \in \mathbb{R}^n$ is the ratios of two nearest neighbors, and d is our intrinsic dimension. $p(\mu|d)$ is the likelihood of the $Pareto(1, d)$ distribution.

$$\prod_{i=1}^{n} \frac{d}{\mu_i^{d+1}} = d^n e^{-(d+1)v}$$

where $v = \sum_{i=1}^{n} ln(\mu_i)$. $p(d) \sim Gamma(a, b)$ is our minimally informative conjugate prior.

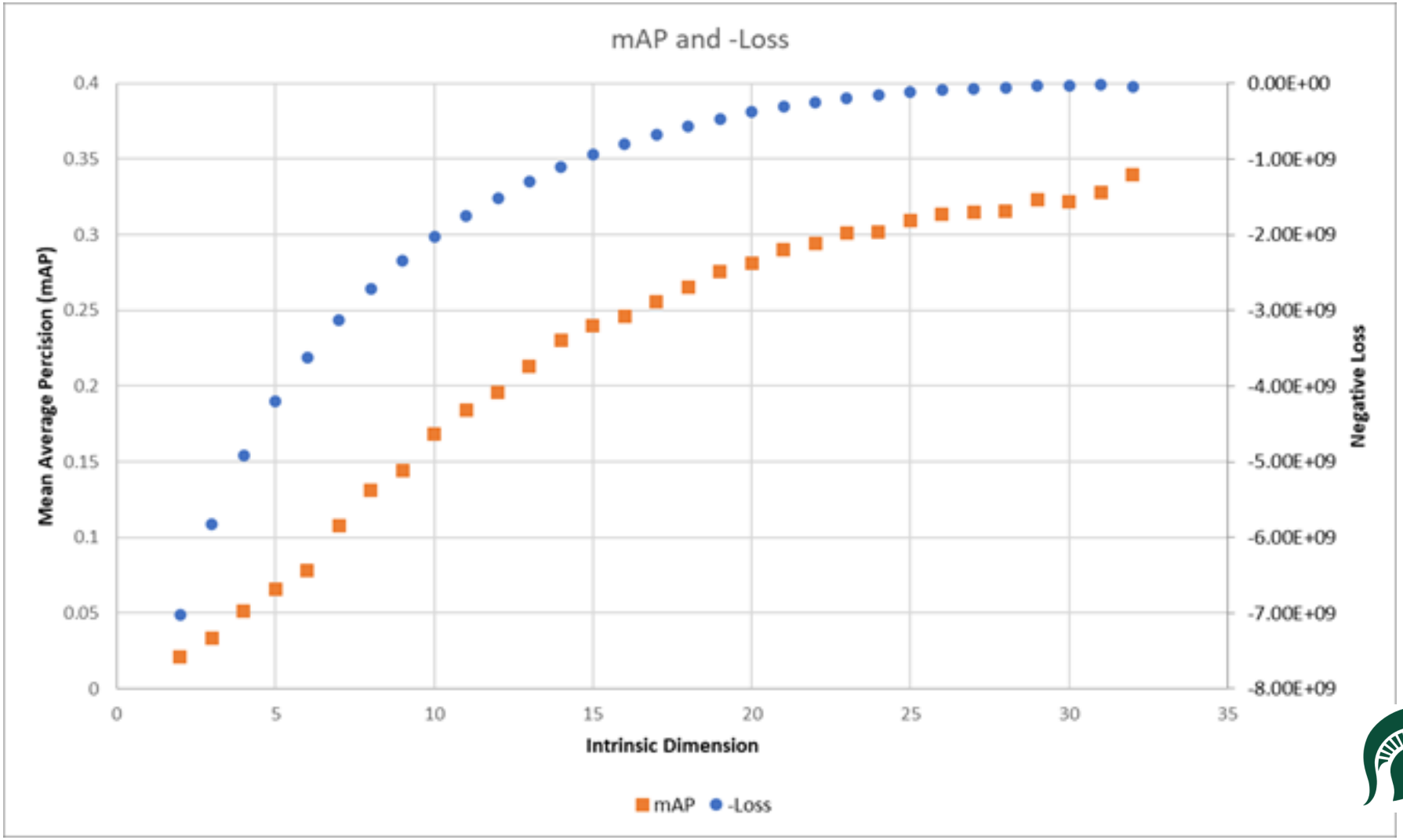
$$p(\mu|d)p(d) = d^n e^{-(d+1)v} \times \frac{b^a}{\Gamma(a)} d^{a-1} e^{-bd} = c d^{n+a-1} e^{-bd-(d+1)v}$$

Where $c = \frac{b^a}{\Gamma(a)} e^{-v}$, which implies that $p(d|\mu) \sim Gamma(a + n, b + v)$

Results

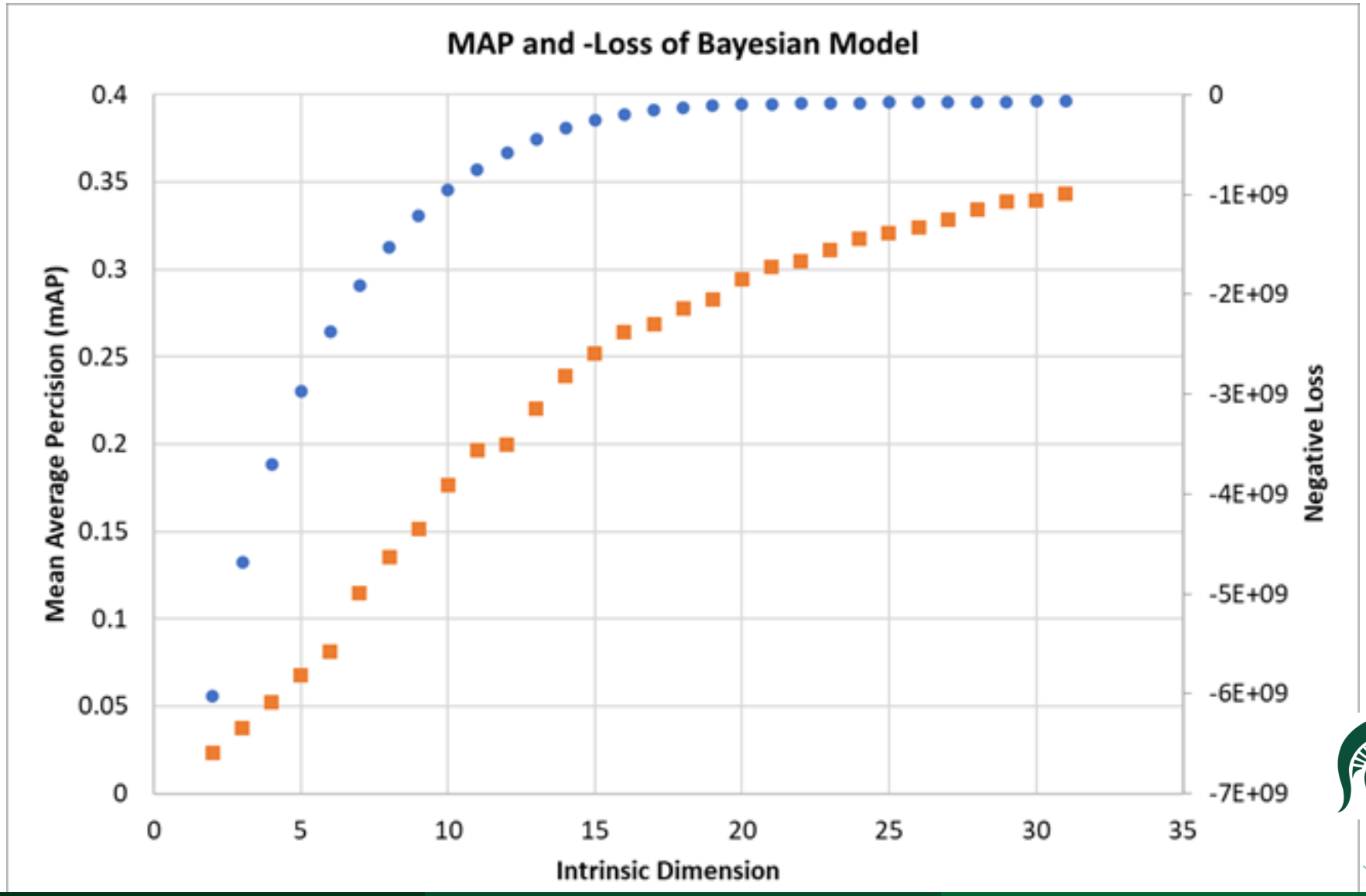
Reproduced Results

Figure: DeepMDS mAP and Loss



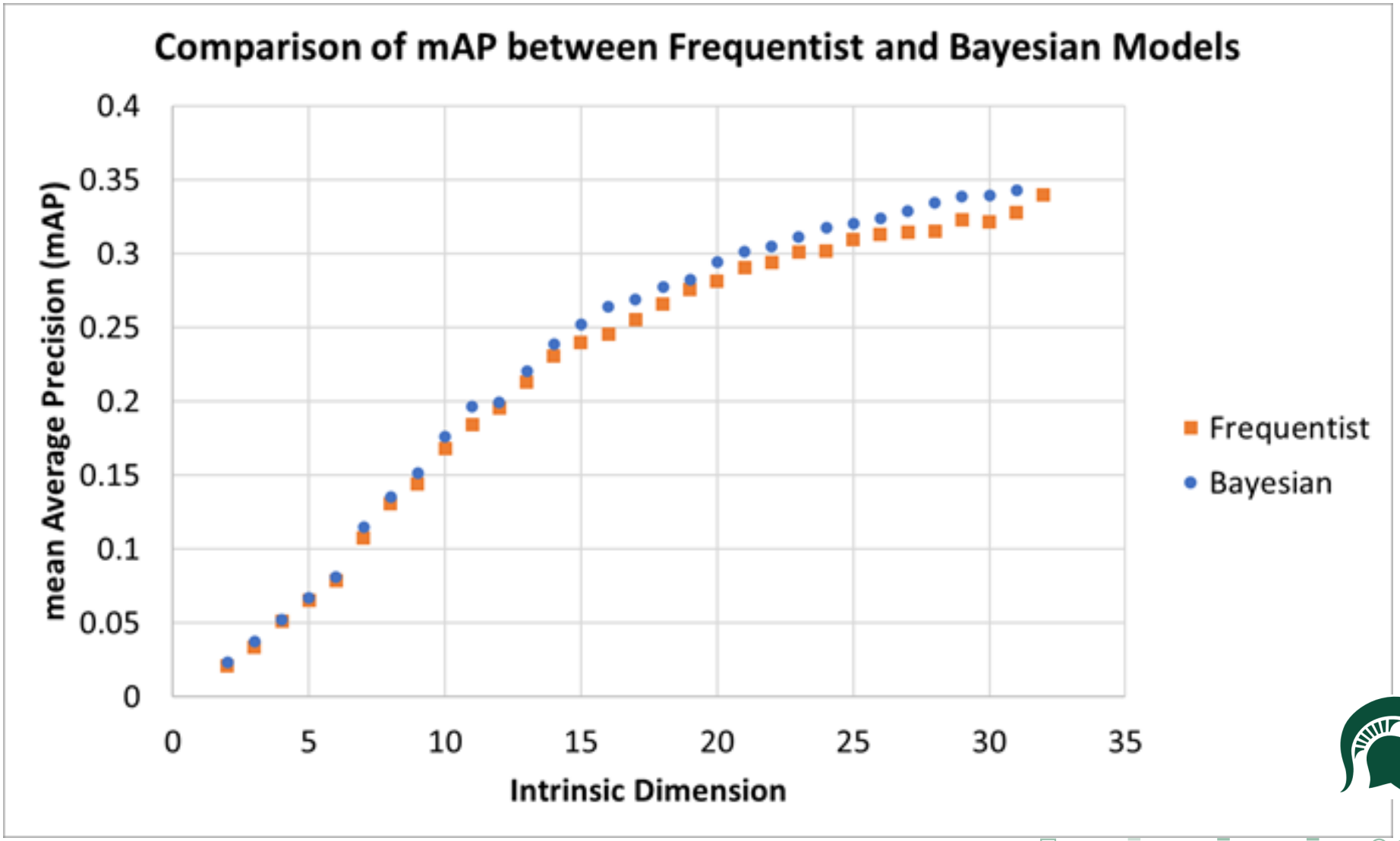
BNN with ImageNet100

Figure: BNN mAP and Loss



Compare BNN vs. DeepMDS

Figure: DeepMDS mAP and Loss



BNN - 64 Intrinsic Dimensions

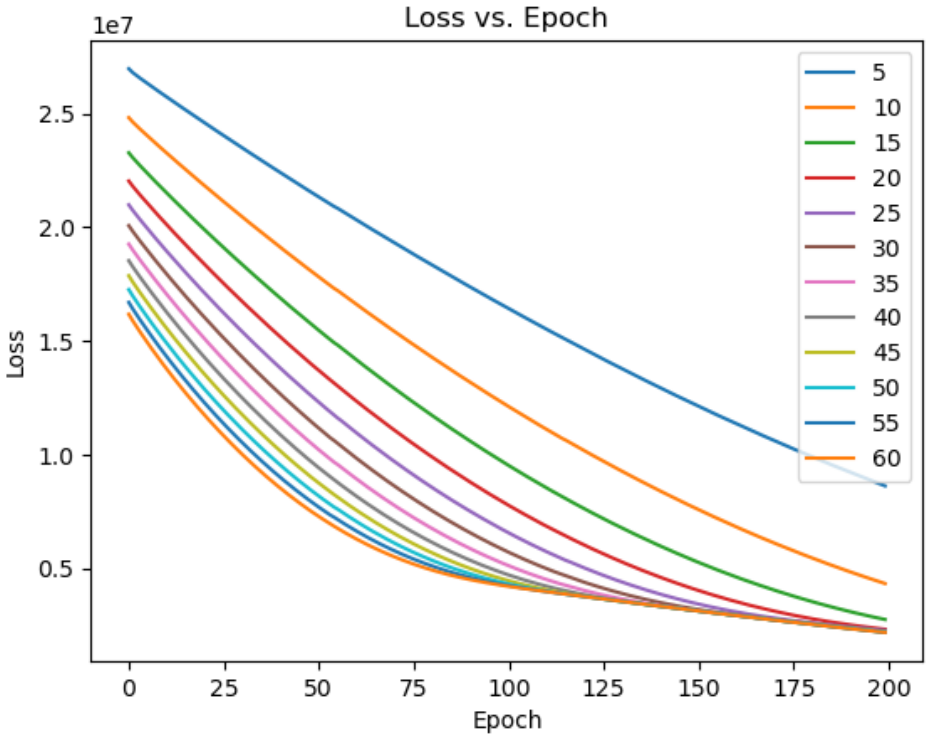


Figure: Loss vs. Epoch

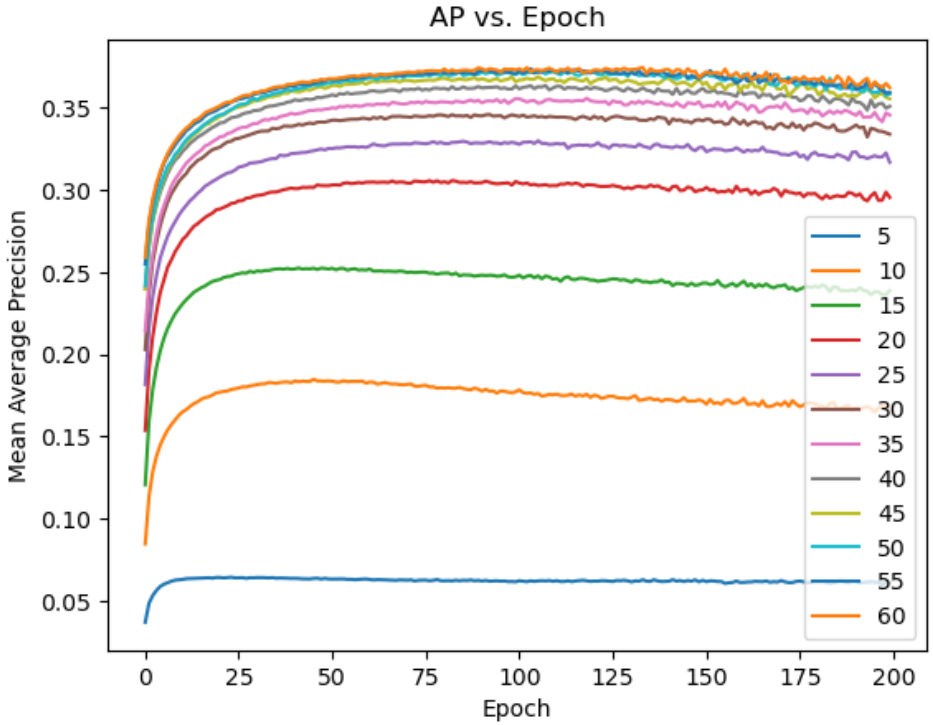


Figure: MAP vs. Epoch

Loss

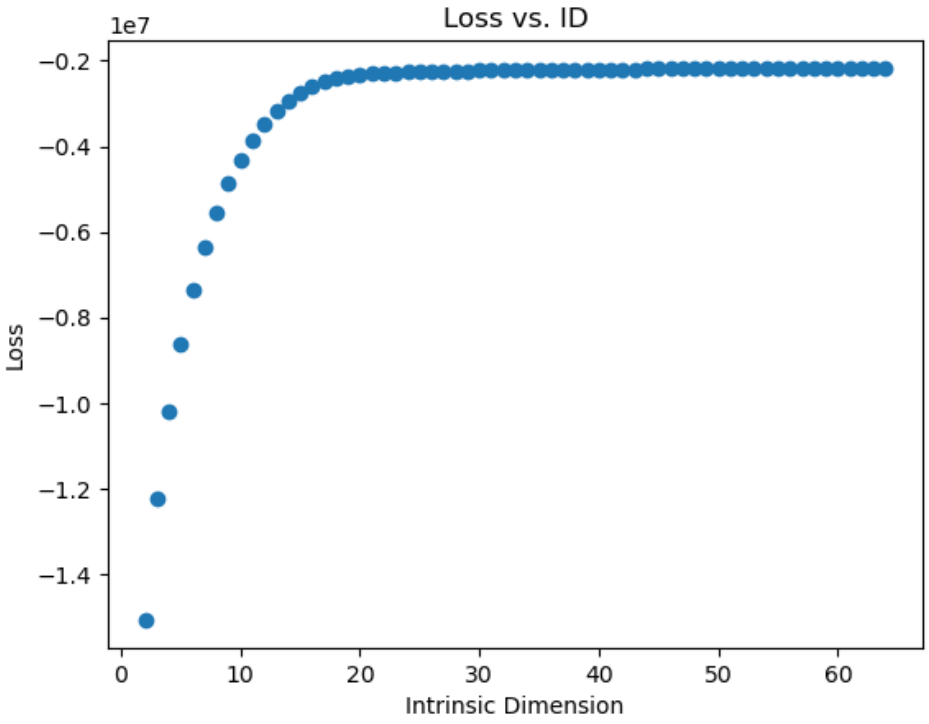


Figure: Loss vs. ID

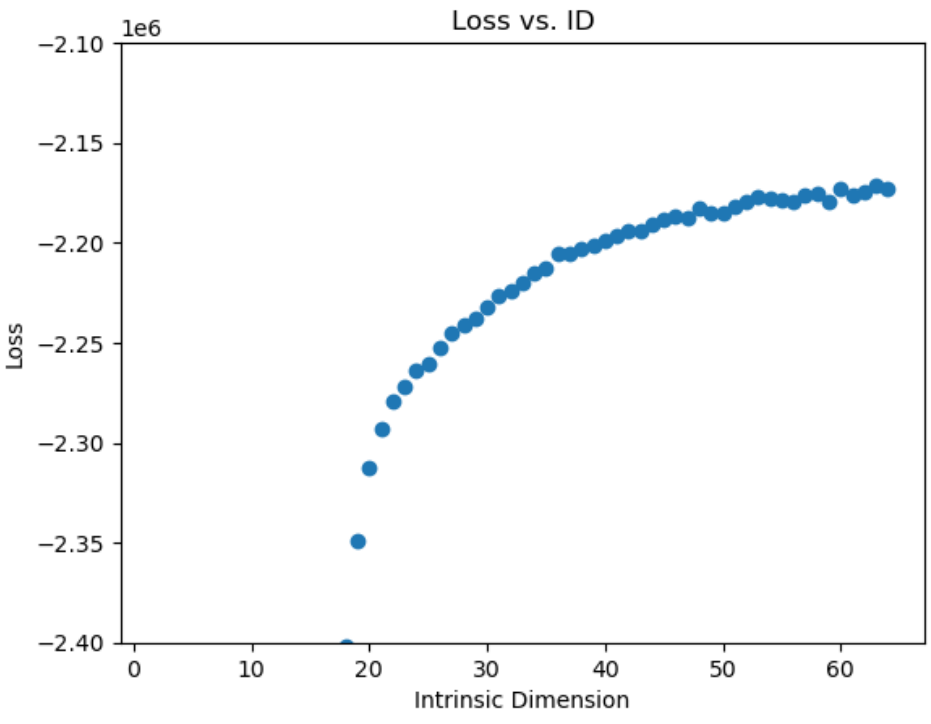
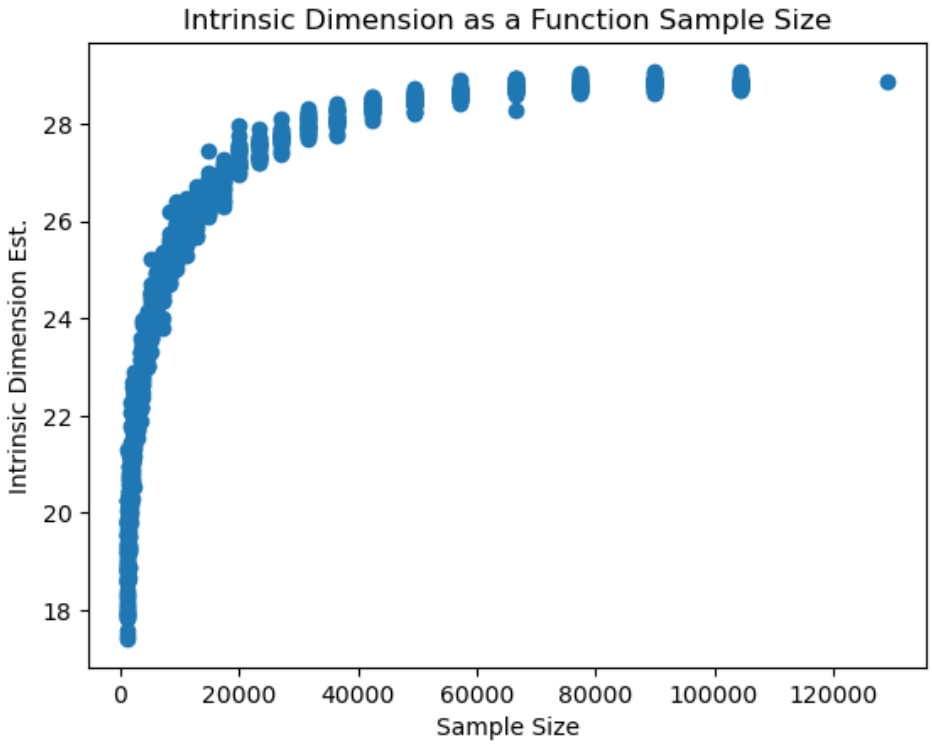
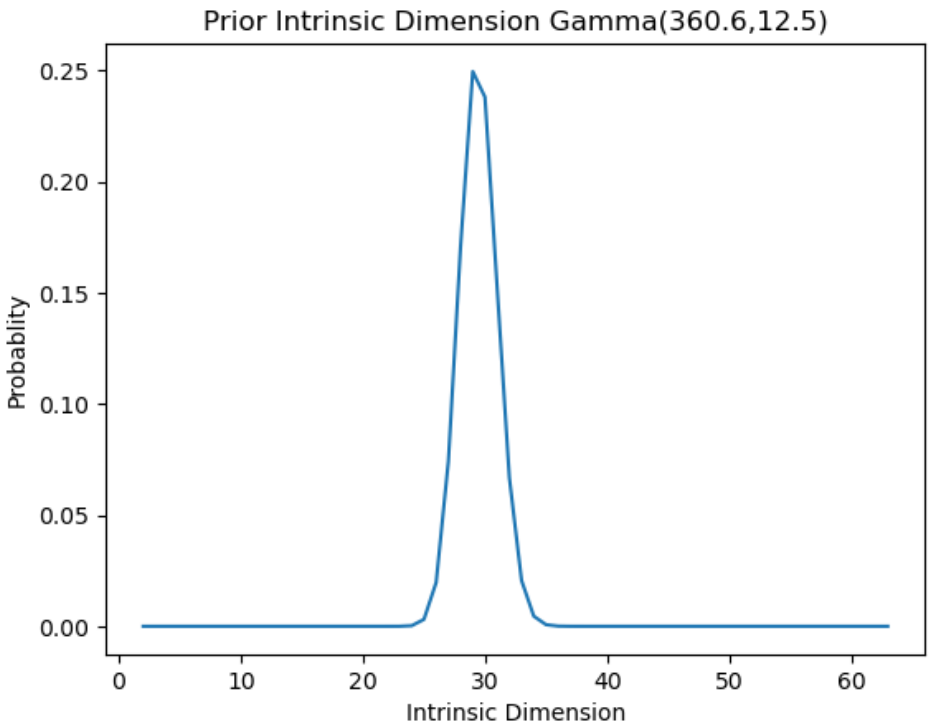


Figure: Loss vs. ID (zoom)

Two Nearest Neighbor Results



Estimated ID based on sample size



ID estimate distribution

Posterior Distribution ID

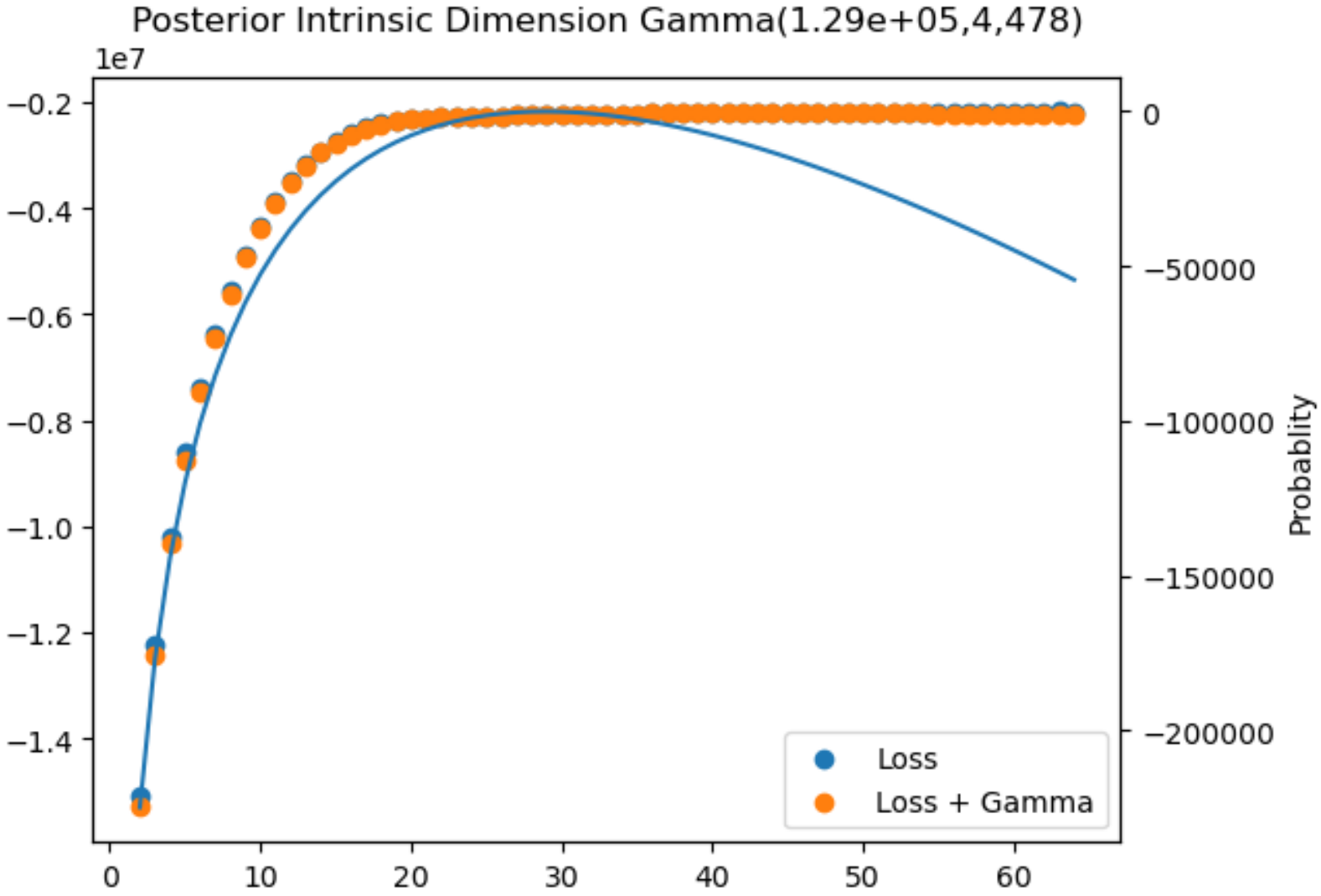


Figure: BNN Loss by Intrinsic Dimension

Conclusions

Conclusions

Rerun of DeepMDS

Reduced from 512 to 19 dimensions
8.8% loss of mAP (36.2% down to 27.4%). 2% more.

Bayesian Neural Network

Reduced from 512 to 2-64 dimensions
At ID=19, 7.7% loss of mAP (36.2% down to 28.5%) 1.1% less.
Estimates an ID=46

Future Work

Future Work

ImageNet 1000

Find ID of ImageNet 1.3 million images.

Transfer Learning

Demonstrate effectiveness of reusing portions of ImageNet100 model to "on board" additional image categories.

Asymptotic Maximum Intrinsic Dimension

Analyze the asymptotic behavior of ID as k categories increase.

References

References

References

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Thank You